

Technical Documentation

Amazon Sales Performance Analysis (2022)

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Tools Used: Microsoft Excel | MySQL | R Programming | Tableau

Date: February 2026

1. Project Overview

This document provides a complete step-by-step record of all data cleaning, transformation, SQL querying, R scripting, and Tableau dashboard creation performed for the Amazon Sales Performance Analysis (2022) project.

2. Dataset Description

Dataset: Amazon Sales Data (2022)

Records: ~30,000+ transactions

Key Columns: Order ID, Category, Ship-State, Status, Amount, Quantity, Date

Currency: Indian Rupees (₹)

3. Data Cleaning & Preparation (Microsoft Excel)

Step 1: Remove Duplicates

Data → Remove Duplicates (Based on Order ID)

Step 2: Handle Missing Values

Used filters to detect blanks in Category, Ship-State, Status, Amount columns.

Step 3: Text Cleaning Formulas

=TRIM(A2) → Remove extra spaces

=PROPER(A2) → Standardize text format

Step 4: Revenue Aggregation & Pivot Tables

=SUM(Amount)

Created Pivot Tables for Category, Ship-State, and Status revenue analysis.

4. SQL Analysis (MySQL)

Total Revenue Query

```
SELECT ROUND(SUM(Amount),2) AS total_revenue  
FROM amazon_sales.`amazon sale report`;
```

Revenue by Category

```
SELECT Category, ROUND(SUM(Amount),2) AS revenue  
FROM amazon_sales.`amazon sale report`
```

```
GROUP BY Category
ORDER BY revenue DESC;
```

Revenue by Ship-State

```
SELECT `ship-state`, ROUND(SUM(Amount),2) AS revenue
FROM amazon_sales.`amazon sale report`
GROUP BY `ship-state`
ORDER BY revenue DESC;
```

Revenue by Order Status

```
SELECT Status, ROUND(SUM(Amount),2) AS revenue
FROM amazon_sales.`amazon sale report`
GROUP BY Status;
```

Cancellation Percentage

```
SELECT ROUND(SUM(CASE WHEN Status = 'Cancelled' THEN 1 ELSE 0 END)
/ COUNT(*) * 100,2) AS cancellation_percentage
FROM amazon_sales.`amazon sale report`;
```

Average Order Value (AOV)

```
SELECT ROUND(SUM(Amount) / COUNT(*),2) AS avg_order_value
FROM amazon_sales.`amazon sale report`;
```

5. R Programming Analysis

Libraries Used:

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(readr)
```

Import Data

```
data <- read.csv('amazon_sales.csv')
```

Revenue by Category (R)

```
category_summary <- data %>%
  group_by(Category) %>%
  summarise(Total_Revenue = sum(Amount))
```

Visualization (R)

```
ggplot(category_summary, aes(x = Category, y = Total_Revenue)) +
  geom_bar(stat = 'identity') +
  theme_minimal()
```

6. Tableau Dashboard Creation

Sheets Created:

- Revenue by Category
- Revenue by Order Status
- Revenue by Ship-State

Dashboard titled 'Amazon Sales Performance Dashboard (2022)' was created by combining the above sheets and formatting currency in ₹.

7. Key KPIs Calculated

- Total Revenue
- Average Order Value (AOV)
- Cancellation Percentage
- Top Revenue Category
- Top Revenue State